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Fracture evolution and failure characteristics of sandstone under freeze-thaw cycling by computed tomography

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ABSTRACT

Long-term freeze-thaw (F-T) action causes the expansion of the fracture network in rock mass with ice-filled joints, triggering more serious engineering geological disasters. This study explores F-T damage effects and loading failure in double-fractured white sandstones with different saturation types (saturated and ice-filled). Saturated intact, saturated fractured, and ice-filled fractured sandstones were subjected to F-T cycle tests and uniaxial compression tests. Three-dimensional images of the specimens under various F-T cycles and stress states were obtained using computed tomography. The evolution in fracture morphology was evaluated visually, with pore network model parameters used for quantitative characterisation. Sandstone fractures evolved according to four main processes: frost-heave cracking, progressive expansion of damage, rock bridge penetration under load, and shear failure. The proportion of connected pores was used to estimate the influences of F-T cycling on pore-structure connectivity and fracture expansion in different specimen layers. During F-T cycling, pore characteristics vary according to size: micropores (equivalent radius < 100 μm) account for >60% of pores and increase in number more than pores of other sizes. Mesopores (100–500 μm) gradually decrease in number but their volume fraction remains >70%. Macropores (500–1000 μm) have the greatest increase in volume fraction. Early F-T cycling induces the connection of pore throats, while later cycling accelerates their expansion. Hydrostatic pressure increases are the leading cause of damage in ice-filled fractured sandstone, which exhibits greater variations in porosity, heterogeneity coefficient, and pore-throat parameters under F-T cycling than saturated fractured sandstone. The results provide insight into fracture evolution and failure characteristics in fractured rock masses with different saturation types (saturated and ice-filled) in cold regions.

1. Introduction

Rock masses are a type of discontinuous medium due to their complex formation processes and environments. They often contain structural planes such as folds, joints and faults, which affect the rock mass in two ways (Cao et al., 2015): by (1) reducing strength and stiffness and (2) becoming a source of secondary structural planes under the actions of weathering, groundwater, external loads and other factors. These secondary structural planes can connect with primary structural planes as they expand, thus further reducing the rock mass's strength and stiffness. Existing research shows that the geometry and arrangement of primary fractures have significant impacts on the strength and deformation of rock masses (Yang et al., 2016), the expansion of secondary fractures (Wang et al., 2020c), and rock bridge fracturing characteristics (Wang et al., 2020b).

In cold regions where the temperature fluctuates periodically near the freezing point of water, freeze-thaw (F-T) weathering is one of the most critical factors leading to the deterioration of rock mass properties (Deprez et al., 2020a). In the past decade, extensive research has been conducted on intact rock in F-T environments, with deterioration characterized by parameters such as porosity, elastic modulus and uniaxial compressive strength (Nicholson and Nicholson, 2000; Jia et al., 2015; Mousavi et al., 2019; Wang et al., 2020a). However, notably, the presence of water in the naturally fractured rock mass will significantly affect its mechanical properties. When the temperature drops below the freezing point, water expands in volume by about 9% due to the water-ice phase transformation. In rock, this causes tension at fracture tips that leads to cracking. Additionally, water can migrate to the frozen zone under the action of ice segregation, which leads to the continuous growth of an ice lens and causes greater frost-heave pressure (Matsuoka

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and Murton, 2008). These two F-T degradation mechanisms lead to broken rock strata, which indirectly lead to geological disasters such as rockfalls and landslides. Therefore, the problem of fracture expansion caused by the frost heave of water has attracted widespread attention. Cao et al. (2020) studied F-T effects on the fracture toughness of semi-circular specimens with different crack inclination angles. Huang et al. (2018) established a mathematical model of frost-heaving pressure in elliptical pores and derived maximum tensile stress and frost-cracking angle equations. Draebing et al. (2017) conducted field monitoring to find that F-T cycling reduces the stability of rock bridges and expands the width of fractures by up to 1 cm. However, these studies have not described the entire process of change in secondary fractures because it is difficult to observe a rock mass's internal structure at the macro-mechanical level. Therefore, the characteristics of secondary fracture evolution need to be quantitatively analysed at the mesoscale.

At present, the methods used to determine pore-fracture structures mainly include acoustic emissions (AE), mercury injection porosimetry (MIP), scanning electron microscopy (SEM), nuclear magnetic resonance (NMR) and X-ray micro-computed tomography (CT; Zhang et al., 2020). However, AE, MIP and SEM have many shortcomings, such as interfering with and damaging samples (Sun et al., 2020). In the T_2 spectra obtained by NMR, it is difficult to observe rock pore structure characteristics directly, and they provide relatively poor microstructural characterisation (Ni et al., 2017). Although CT technology generally has an insufficient spatial resolution to identify nanoscale micropores, Deprez et al. (2020a) believe that it is adequate for studying fracture evolution characteristics and has broad application prospects in assessing F-T damage. De Kock et al. (2015) used freezing/micro-CT equipment to observe crack growth and ice crystallisation exotherms during freezing and subsequent closing of cracks during thawing. Shields et al. (2018) combined CT with AE to study cracking behaviour in mortar specimens at different saturations during F-T. Liu et al. (2019) performed CT scans on sandstones that underwent a single F-T cycle at different confining pressures. They found that the rate of increase in porosity has a linear relationship with confining pressure. Maji and Murton (2020) studied microscopic structural changes in sandstone and limestone in the early stage of F-T using statistical methods. The results showed that a modified lineal-path function can effectively quantify F-T fracture characteristics. Although CT technology has provided valuable mesoscale data in intact rock, there are few reports on F-T damage characteristics in fractured rock.

On the other hand, most existing work has investigated F-T effects in saturated rock; however, McGreevy and Whalley (1985) noted that the influence of moisture distribution should also be considered. The damage caused by saturated F-T (100% saturation, but there is no water inside the macro-fractures) cannot represent the frost heave of fractures caused by ice-filled F-T (with water-filled fractures after thawing and ice-filled fractures after freezing). This is because there is an essential difference between the average water content of rocks and the water content of fractures. Existing studies have shown that day and night or seasonal temperature changes in cold regions, such as high mountainous areas, will make slopes with ice-filled rock joints unstable (Matsuoka and Murton, 2008; Mamot et al., 2018). This will significantly affect the route selection design of roads or railways and pose a potential threat to their construction and safe operation. Therefore, a further comparative study on the difference between saturated and ice-filled F-T can provide a reference for the correct evaluation and prediction of engineering geological disasters, such as landslides.

The above review shows that existing studies mainly focus on the mechanical properties and meso-structural deterioration in intact rocks. In contrast, damage evolution and failure characteristics in fractured rocks with different saturation types (saturated and ice-filled) in the F-T environment remain unclear. Accordingly, the present study subjected saturated intact specimens, saturated fractured specimens and ice-filled fractured specimens to F-T cycle tests and uniaxial compression tests. CT technology was used together with Avizo 3D visualisation software to study the process of secondary fracture evolution under various F-T cycles and stress states. The meso-parameters obtained are of great significance for an in-depth understanding of rock damage characteristics under different saturation types in cold regions.

2. Methodology

2.1. Preparation of specimens

The experiments used white sandstone obtained from Zigong City, Sichuan Province, China. According to the requirements of International Society for Rock Mechanics (ISRM), standard intact rock samples of ∅ 50 mm × 100 mm were fabricated. Previous research shows that rock slopes often form steep-slow fractures approximately parallel to the slope surface that are connected by a complete rock bridge (Huang et al., 2015). The rock mass may slip along the fracture surface under weathering and external loading, causing “step-path” failure. To replicate the steep-slow fractures, rock samples with intermittent double-fractures were fabricated by drilling and wire cutting. With consideration of engineering data, we selected fracture inclination angles of 15°–75°, a fracture length × width of 20 mm × 1 mm, and a rock bridge length of 15 mm. The geometry of the fractures is shown in Fig. 1.

The rock samples were baked in an oven at 105 °C for 48 h, then saturated with a vacuum saturation device after cooling. When the vacuum reached −0.1 MPa, the rock samples were kept under this condition for 24 h to ensure complete saturation. Two prefabricated double-fractured rock samples were selected for saturated and ice-filled F-T testing according to their dry density and P-wave velocity. Another intact specimen was used in a saturated F-T test to determine the influence of prefabricated fractures on the expansion of secondary fractures. The sample numbers and basic parameters are shown in Table 1.

2.2. Experimental equipment and process

This experiment was mainly composed of F-T cycle tests (Part 1) and uniaxial compression tests (Part 2), both of which were followed by CT scanning. The experimental flowchart is shown in Fig. 2. In this flowchart, i_A, i_B and i_C represent the numbers of F-T cycles applied to Specimens A, B and C, respectively. The CT equipment (Multiscale-Voxel series industrial CT machine, Tianjing Sanying Precision Instruments Co., Ltd.) used scanning parameters of: current = 1.3 mμA, voltage = 385 kV, exposure time = 0.5 s and spatial resolution = 41.67 μm. Each specimen was scanned five times after being subjected to various F-T cycle and stress state conditions (Fig. 2).

The operational details of the F-T cycle and uniaxial compression tests were as follows.

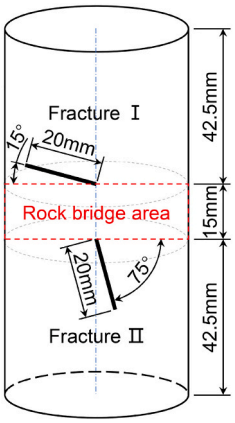


Fig. 1. Geometry of the fracture space.

Table 1
Sample types and their basic parameters.

Sample number	Sample type	Dry density (g/cm^3)	Porosity (%)	P-wave velocity (m/s)
A	Saturated intact rock	2.323	10.25	2986
B	Saturated fractured rock	2.266	10.75	2577
C	Ice-filled fractured rock	2.275	10.98	2629

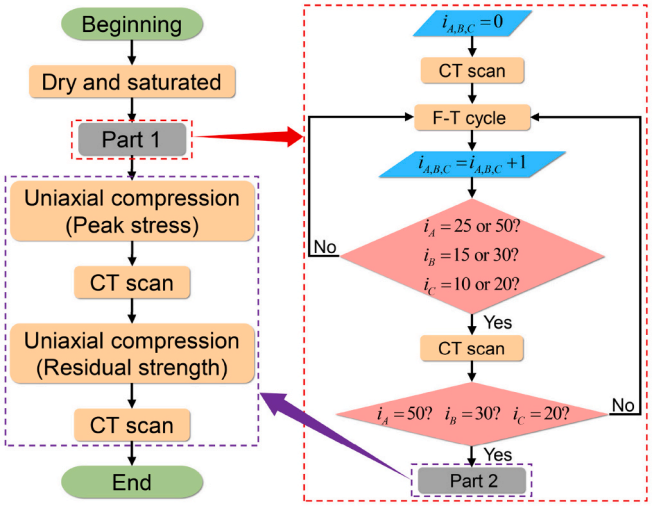


Fig. 2. Experimental procedure.

(1) F-T cycle tests: Specimens A and B were wrapped with film to retain moisture. Specimen C was placed into an 80 mm × 130 mm rectangular plastic bag filled with water until the specimen's top surface was just covered and ensure no bubbles in the fractures to achieve 100% fracture saturation. In a low-temperature environment, water gradually freezes and forms ice-filled fractures. The effect of water injection and freezing is shown in Fig. 3. Because the plastic bag was malleable and open at the top, it did not produce frost-heaving pressure on the specimen's surface (Huang et al., 2019). To conduct one F-T cycle, the specimen was frozen at −20 °C for 8 h, then taken out of the bag and thawed in a water bath at 20 °C (± 2 °C) for 8 h to ensure complete saturation. According to the changes in P-wave velocity and surface particle exfoliation of pre-test, Specimens A, B and C should be subjected to 50, 30 and 20 F-T cycles, respectively.

(2) Uniaxial compression tests: A rock mechanics testing machine (TAW-1000, Changchun City Chaoyang Test Instruments Co., Ltd.) in an environment of 20 °C was used with a loading rate of 0.05 mm/min. Since the specimens did not produce obvious secondary fractures within the elastic deformation range (Ju et al., 2018), they were directly loaded to the first stage's peak stress state and then unloaded for CT scanning. In the second stage, specimens were reloaded to the residual intensity stage, then unloaded and a CT scan performed.

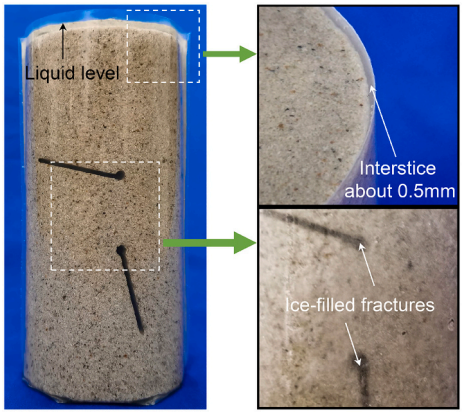


Fig. 3. Water injection and freezing effects in an ice-filled specimen.

2.3. 2D image processing

As the inherent defects of the CT equipment can cause image noise and artefacts (Fig. 4(a)) that interfere with fracture extraction, the first step was to reduce noise in the original images. Although the traditional median filter can eliminate "salt-and-pepper"-type noise, it has a poor effect on ring artefacts. The *Non-Local Means* algorithm can denoise all pixels in an image through weighted averaging, effectively reducing both Gaussian noise and ring artefacts. A slice image processed by *Non-Local Means* filtering is shown in Fig. 4(b).

Different substances have different X-ray absorption capacities that result in different grey values in an image. Substances with greater densities (such as quartz) cause higher grey values and pixel brightness, resulting in a white image. In contrast, lower-density features like the rock matrix and fractures appear grey and black, respectively. Fig. 4(c) shows the grey value distribution of the detection line, in which A, B and C represent minerals, a secondary fracture and a prefabricated fracture, respectively.

The 2D image threshold segmentation process used has a substantial impact on 3D rock fracture reconstruction. Different segmentation thresholds and methods will result in different porosity and pore structure data. Commonly used methods include the *Watershed Separation* and *Interactive Thresholding* processes in Avizo software. *Watershed Separation* uses a semi-automatic method to define pores, which is unsuitable for processing materials with complex pore structures such as sandstone. Therefore, *Interactive Thresholding* was used to separate the pore structures, with the *Interactive Top-Hat* algorithm used for optimization. The latter can identify microfractures with no obvious grey differences by using luminance differences. In addition, because the greyscale histograms of sandstones usually contain a single peak, the segmentation threshold cannot be obtained directly according to changes in grey level. Therefore, referring to the work of Peng et al. (2011) and based on the porosity measured during the saturation experiment prior to F-T, the inverse analysis method was used to determine the segmentation thresholds for Specimens A, B and C.

3. Fracture evolution characteristics

3.1. Fracture morphology evolution

A more intuitive evaluation of fracture morphology evolution in specimens subjected to various F-T cycles and stress states can be obtained via 3D reconstruction. To avoid edge artefacts, we used ∅ 40 mm × 60 mm cylindrical rock cores (Fig. 5). Image cropping was done using the *Register Images* command in Avizo software to ensure that the cropped areas were of the same spatial position regardless of F-T cycling and stress states. Also, to highlight the main changes, we adjusted the threshold using the *Interactive Top-Hat* command to omit the original connected pore structure and only show the parts with a volume > 10 mm^3. The process fracture morphology evolution is shown in Fig. 6.

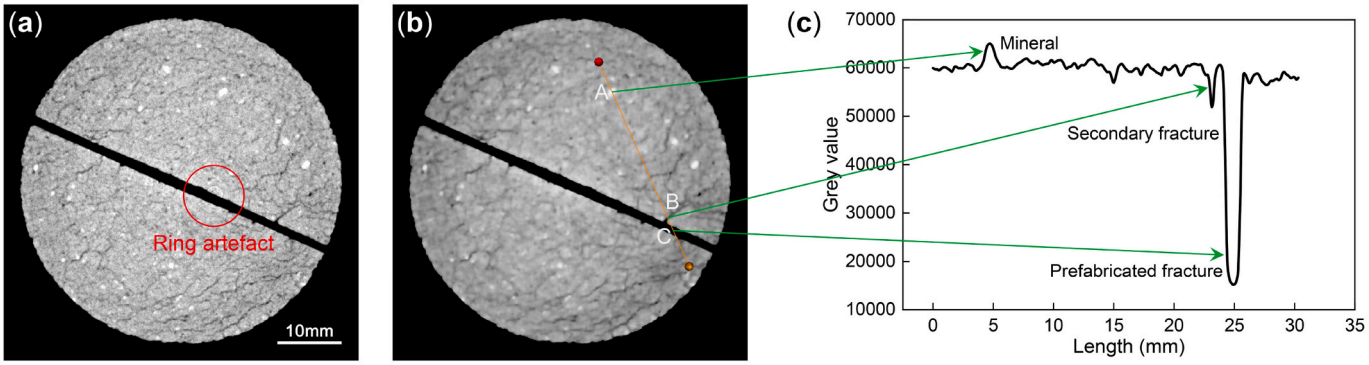


Fig. 4. Specimen C after 20 F-T cycles: Original image of a slice (a), image after non-local means filtering (b) and grey-value distribution (c).

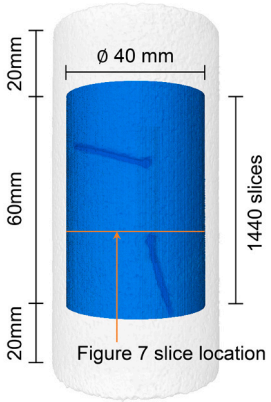


Fig. 5. Sample core extraction.

As shown in Fig. 6(a), Specimen A had a uniform texture before F-T with no apparent fractures. After 25 F-T cycles, flaky vertical fractures appeared at the edge and the total fracture volume (V_F) was 54 mm^3. After 50 F-T cycles, concentric annular fractures were generated, fractures developed from both ends to the middle, and V_F increased to 229 mm^3. In the process of uniaxial compression, the friction between the end-face of the specimen and the metal cushion of the testing device put the two ends of the specimen in a 3D stress state. This caused the primary fracture to expand from the middle to the two ends, which led to the closure of some edge fractures. In the peak stress state, the primary fracture did not completely penetrated the specimen and $V_F = 470$ mm^3. In the residual strength stage, the primary fracture was further connected with edge fractures, exhibited typical shear failure, and V_F increased sharply to 1050 mm^3.

Fig. 6(b) shows a few microfractures along the edges of prefabricated fractures in Specimen B after 15 F-T cycles. At 30 F-T cycles, the microfractures gradually expand and produce flaky vertical fractures similar to those in Specimen A. In the peak stress state, the prefabricated fracture penetrates from the end to alongside the rock bridge area. The upper end of Fracture I is close to the edge of the specimen and the F-T damage is more serious, so a vertical macro-fracture is formed before the lower end of Fracture II. Additionally, a shear crack occurs in the middle of Fracture I. Simultaneously, Fracture II's inclination is similar to that of the shear fracture plane, so there is no secondary crack in the middle. The main things to note are the continuous expansion of the lower end of Fracture II and a penetrating fracture in the rock bridge area in the residual strength stage. After shear failure, Specimen B's fracture morphology is more complicated than that of Specimen A.

As shown in Fig. 6(c), frost-heave expansion occurs in the middle of the prefabricated fractures in Specimen C after 10 F-T cycles. At 20 F-T cycles, the local damage caused by frost-heaving pressure gradually expands and many irregular microfractures appear. The disappearance of frost-heave fractures in the central area may be caused by the closure of some fractures or errors in the threshold segmentation process. By comparing the 2D slices shown in Fig. 7(a) and (c), it can be seen that the frost-heave fractures in Specimen B are more concentrated, while those in Specimen C are widely distributed inside the rock and the damage in the central area is more serious. Under load, Specimens C and B have similar failure modes; however, because of the greater deterioration in Specimen C, the volume of its primary fracture is greater. Also, comparison of Fig. 7(b) and (d) shows that the fracture-closure phenomenon is more extensive in Specimen C.

In summary, in fractured rock that becomes damaged under load after F-T cycling, the fracture evolution involves four processes: frost-heave cracking, progressive expansion of damage, rock bridge penetration under load, and shear failure.

The total volume (V_F) and total surface area (A_F) of fractures are closely related to the characteristics of fracture morphology evolution. The trends in these two parameters at different CT scanning times are shown in Fig. 8(a). Furthermore, Table 2 shows the rates of increase in V_F and A_F at two CT scan instances in Specimens B and C. Under the action of F-T cycling, both V_F and A_F showed a trend of accelerating increase. Compared with saturated F-T conditions, ice-filled F-T can cause greater expansion damage in fractured rock. However, since ice-filled F-T causes more serious damage, the effect of uniaxial compression on crack propagation is more significant in Specimen B than Specimen C, and its rates of increase in V_F and A_F are greater.

The specific surface area of fractures (S_{VF}), defined as the ratio of total surface area to total volume (Eq. (1)), is used to evaluate the fracture structure's complexity. It represents the total area of a fracture per unit volume.

$$S_{VF} = \frac{A_F}{V_F} \tag{1}$$

Changes in the specific surface area of fractures in Specimens B and C are shown in Fig. 8(b). It shows that S_{VF} peaks at the point of peak stress; in the residual strength stage, S_{VF} decreases due to compaction and closure of secondary fractures. The difference between Specimens B and C is that the S_{VF} of Specimen B increases faster before the peak stress point, with an increase of 36.32%. In contrast, the S_{VF} of Specimen C increases sharply in the later stage of F-T, with an increase of 44.40%, while the increase before the peak stress point is only 11.88%. These data show that ice-filled F-T can easily cause secondary fractures and reduce the engineering stability of rock masses.

3.2. Fracture evolution analysis

The rock bridge area formed by combinations of double fractures has an essential influence on the engineering stability of rock masses. This area is more prone than others to frost-heave damage in low-temperature

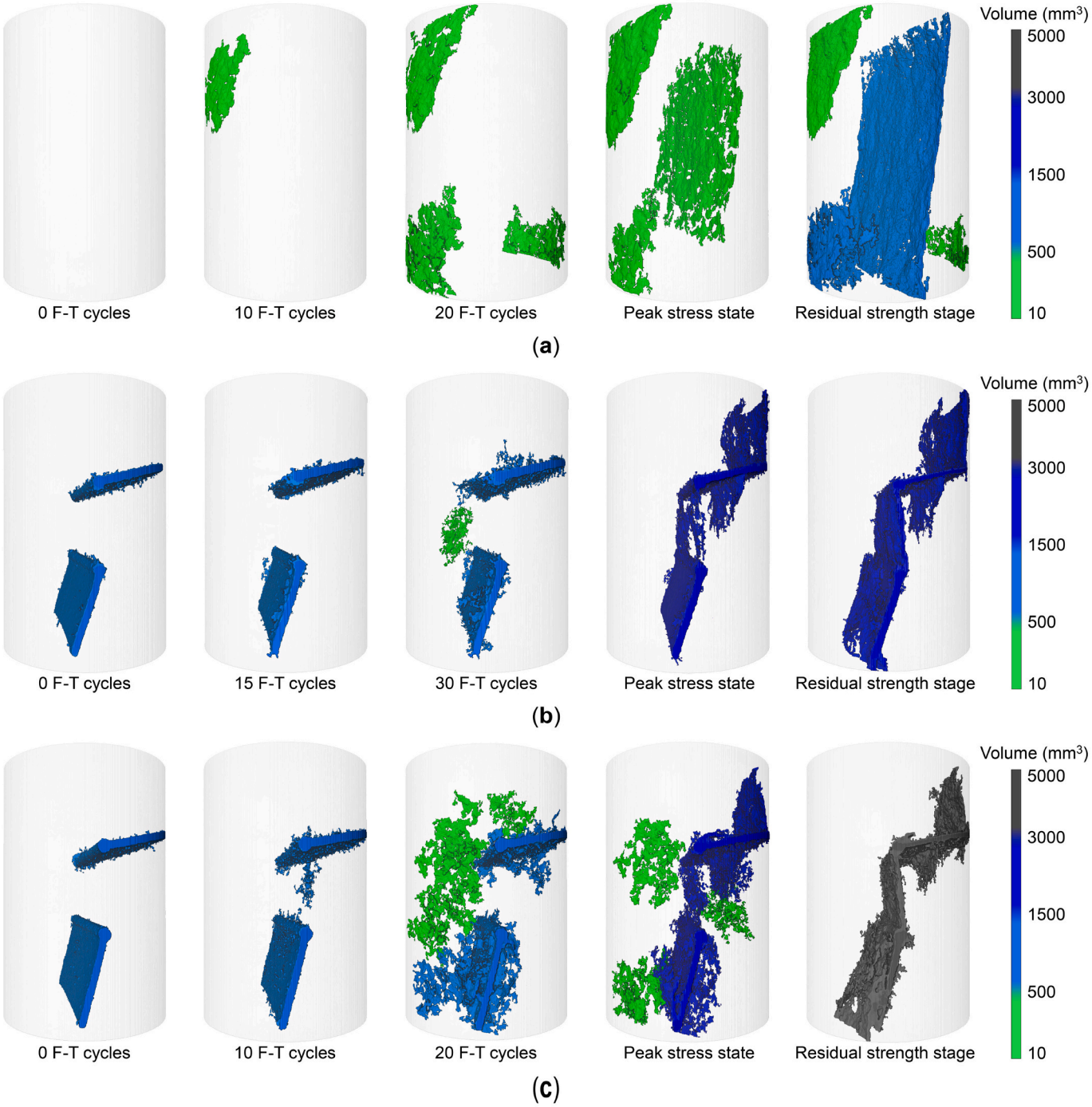


Fig. 6. 3D reconstructions showing the evolutionary process of fracture morphology in Specimens A (a), B (b) and C (c).

environments. To investigate this, we analysed a cylindrical core of a rock bridge area (diameter × height = 350 × 350 voxels; Fig. 9).

3.2.1. Porosity change

The effect of F-T cycling on the connectivity of the fracture structure was evaluated by analysing changes in specimen porosity. Table 3 summarises the connected porosity P_{con} (volume of connected pores as a percentage of the core volume), total porosity P_{tot} (total pore volume as a percentage of the core volume) and proportion of connected pores P_R (P_{con} / P_{tot}). The table shows increases in parameter P_{tot} in response to F-T cycling; specifically, the increases are: Specimen A = 4.00% and 11.54%, Specimen B = 3.26% and 12.16%, and Specimen C = 9.74% and 21.91%, respectively. In contrast, P_R shows the opposite trend, with increases of: Specimen A = 98.15% and 36.83%, Specimen B = 86.69% and 28.17%, and Specimen C = 112.18% and 64.03%, respectively.

According to the 2D image on the *x*-*y* planes, the specimen's areal porosity along the *z*-axis can be obtained and used to explore fracture expansion differences in different layers under different saturation modes. The variations in connected areal porosity S_{con} (percentage of the total slice area occupied by connected pores), total areal porosity S_{tot} (percentage of the total slice area occupied by all pores), and proportion of 2D connected pores S_R (S_{con} / S_{tot}) are shown in Fig. 10. Additionally, to evaluate heterogeneity in the fracture distributions, a heterogeneity coefficient (U_z) was calculated (Eq. (2); Fan et al., 2018). It is defined as the ratio of the standard deviation to the average value, reflecting the degree of dispersion of areal porosity.

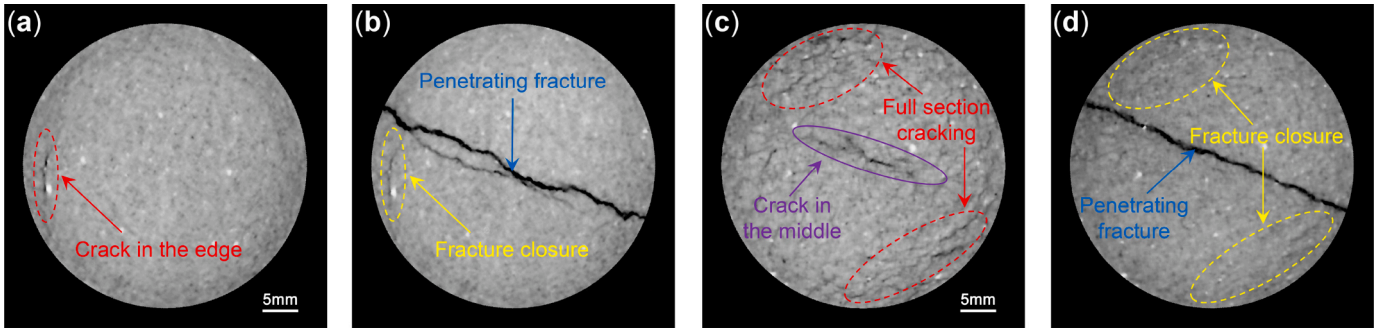


Fig. 7. 2D morphology of specimen fractures. Specimen B after 30 F-T cycles (a) and in the residual strength stage (b); Specimen C after 20 F-T cycles (c) and in the residual strength stage (d).

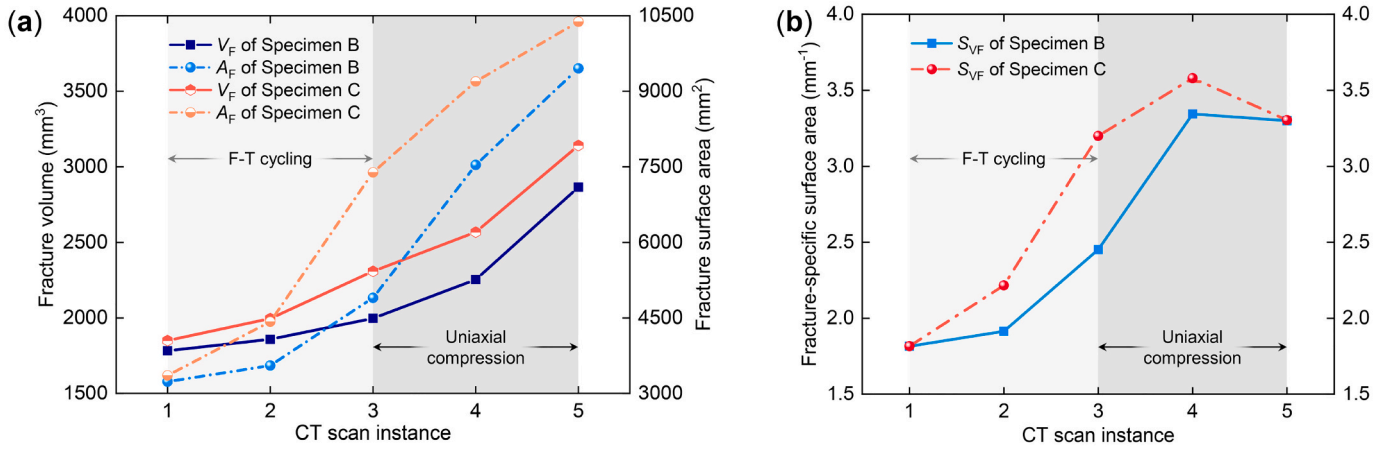


Fig. 8. Variations in fracture volume (V_F) and surface area (A_F) (a) and specific surface area of fractures (S_{VF}) (b) according to CT scan instance in Specimens B and C.

Table 2
Rates of Increase in V_F and A_F under F-T cycling and uniaxial compression.

Parameter	F-T cycling		Uniaxial compression	
	R_1 (%)	R_2 (%)	R_3 (%)	R_4 (%)
V_F, Specimen B	4.20	7.48	12.81	27.12
V_F, Specimen C	7.89	15.68	11.21	22.33
A_F, Specimen B	9.86	37.65	53.84	25.41
A_F, Specimen C	31.63	67.02	24.42	12.89

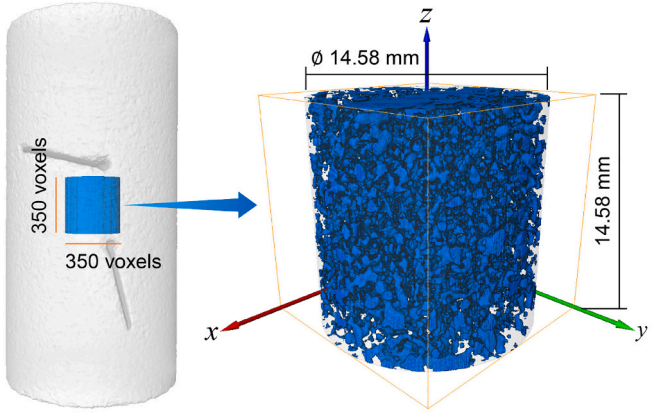


Fig. 9. Core extracted from a rock bridge area.

$$U_z = \frac{1}{\overline{S}_z}\sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(S_{tot} - \overline{S}_z\right)^2} \tag{2}$$

Where $\overline{S}_z = \frac{1}{n}\sum_{j=1}^{n} S_{tot}$ denotes the mean value of S_{tot}; n is the number of CT images used for the analysis.

Fig. 10(a)–(c) show that during the F-T cycling of Specimen A, the S_R of each 2D slice increases uniformly and the amplitude of fluctuation in the curve gradually becomes stable. Parameter U_z shows a decreasing trend, with values of 13.19%, 12.01% and 10.93% at 0, 25 and 50 F-T cycles, respectively. Fig. 10(e)–(i) show that, due to the prefabricated fractures, the areal porosities at both ends of Specimens B and C are more extensive than in the middle sections. Although this is clearly different to the case of Specimen A, this has important significance in studying the influence of primary fractures on surrounding secondary fractures in an F-T environment. During the F-T process in a fractured specimen, S_R increased from the vicinity of the prefabricated fracture and gradually expanded to the centre of the rock bridge area. The amplitude of fluctuation in the curve increases in the early F-T period and decreases in the late period. Accordingly, Specimen B's U_z values were 23.78%, 37.04% and 35.59% after 0, 15 and 30 F-T cycles, respectively, exhibiting a process of rapid increase and slow decrease. The U_z values of Specimen C were 26.91%, 30.23% and 16.85% after 0, 15 and 30 F-T cycles, respectively, indicating a sudden drop in the late period of F-T cycling. After F-T cycling in Specimen C, the U_z value obtained from the 300 slices in the centre of Fig. 10(i) (slices 25 to 325) was only 6.56%.

Table 3
Changes in porosity parameters (%) in Specimens A, B and C after different numbers of F-T cycles.

Parameter	Specimen A			Specimen B			Specimen C		
	0 cycles	25 cycles	50 cycles	0 cycles	15 cycles	30 cycles	0 cycles	10 cycles	20 cycles
P_{con}	2.56	5.26	8.02	2.76	5.56	7.91	2.85	6.85	13.27
P_{tot}	10.25	10.66	11.89	10.76	11.10	12.45	10.98	12.05	14.69
P_R	24.91	49.36	67.54	25.16	46.97	60.20	25.94	55.04	90.28

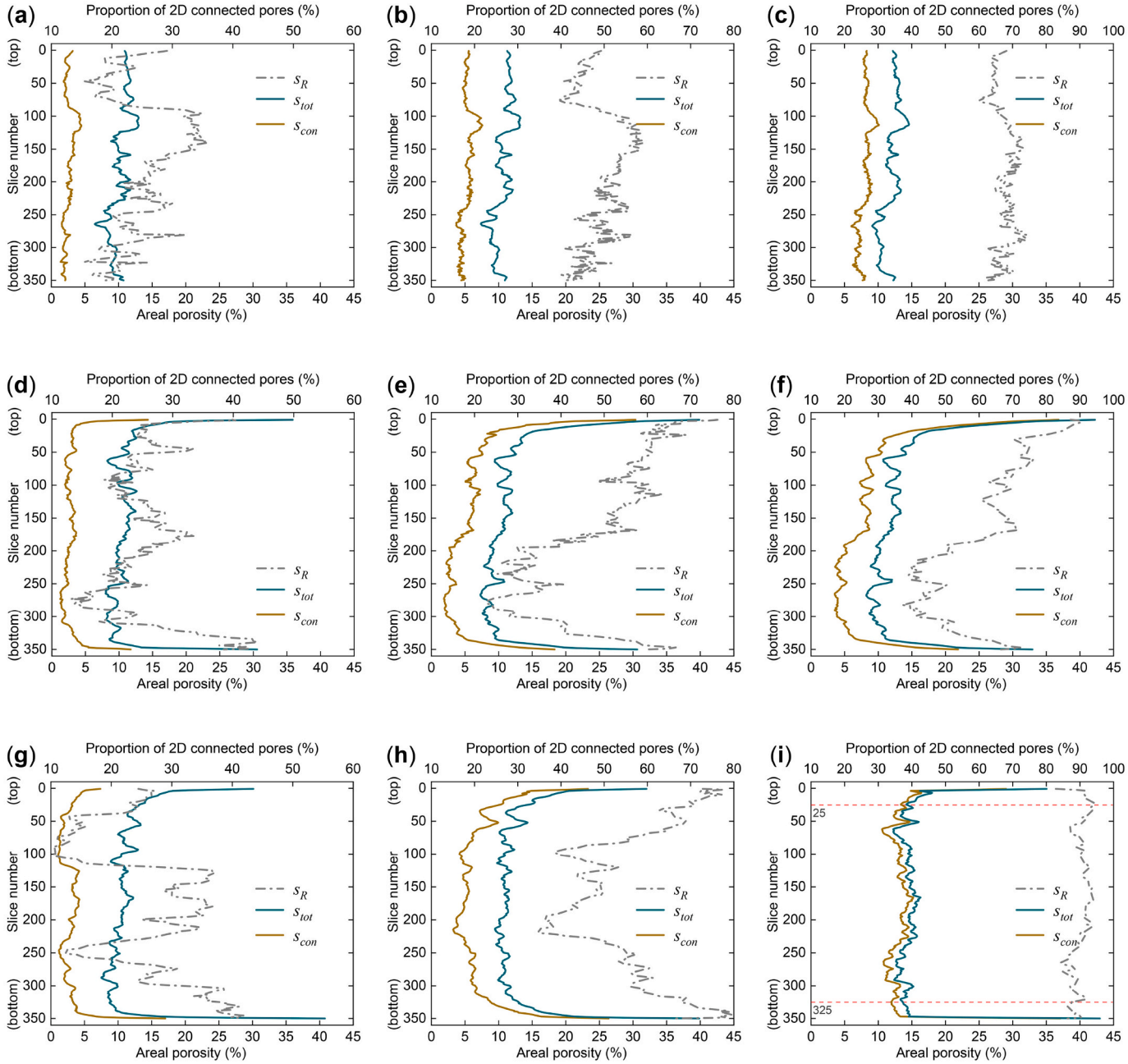


Fig. 10. Variations in areal porosity with slice number after various numbers of F-T cycles in each specimen. Specimen A after 0 (a), 25 (b) and 50 cycles (c); Specimen B after 0 (d), 15 (e) and 30 cycles (f); and Specimen C after 0 (g), 10 (h) and 20 cycles (i).

3.2.2. *Pore network model parameters*

To more accurately reflect the spatial distributions of the pore-fracture structures, a pore network model (PNM) was used to quantitatively analyse the samples in different states. In the early use of this model, Lindquist et al. (1996) extracted the 3D structure of sandstone based on skeletonization using the medial axis transform. In addition, Baldwin et al. (1996) and Silin et al. (2003) proposed the watershed and maximal ball methods, respectively. Fig. 11 shows a typical pore-throat-pore geometry. The sphere represents the pores, and the channel represents the throat. In this paper, based on the *Pore Network Model* of Avizo, a simplified topological structure was established using the medial axis algorithm (Isaka et al., 2019). Then, the fracture morphology was

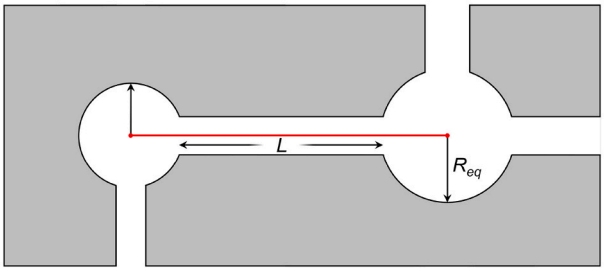


Fig. 11. Schematic diagram of a typical pore-throat-pore conduit, which was modified from Zhang et al. (2020).

characterized in terms of the pore (throat) equivalent radius, throat length and other parameters. Assuming that the pores are equivalent to spheres of the same volume, the equivalent radius can be determined as:

$$R_{eq} = \sqrt[3]{\frac{3V}{4\pi}} \tag{3}$$

Where R_{eq} represents the equivalent radius (μm) and V is the pore (or throat) volume (μm^3).

A 3D pore-throat model was constructed (Fig. 12). It shows that the overall pore-throat structures of saturated specimens remain similar under F-T cycling. However, the larger pores are mainly distributed on the outer edges of Specimen A, while they are concentrated near the prefabricated fractures in Specimen B, which is similar to the result shown in Fig. 6. On the other hand, ice-filled Specimen C presents a more complex pore-throat structure with a denser distribution.

Analysis of the pore-throat parameters shows that saturated F-T Specimens A and B have similar changes, so the following analysis only compares fractured specimens according to different F-T modes. The pore-throat parameters of Specimens B and C are given in Tables 4 and 5, respectively. The tables show that the pore number, average volume, and the number of throats in Specimen B increase by 1.98%, 1.06%, and 12.24%, respectively, during the early F-T period, then increase by 3.34%, 8.59%, and 10.04%, respectively, in the late period. In Specimen C, these parameters increased by 4.16%, 5.43%, and 56.79% in the early period and by 4.70%, 16.22%, and 52.93% in the late period. The main difference is that ice-filled F-T cycling dramatically increases the development of throats. After uniaxial compression, thenumbers of pores and throats in Specimen B decrease by 11.99% and 38.13%, respectively, and the average volume of pores increases by 69.83% compared to before loading. The numbers of pores and throats in Specimen C decrease by 21.29% and 50.80%, respectively, and the average volume of pores increases by 81.31%.

The pore equivalent radius distributions of Specimens B and C under different numbers of F-T cycles and different stress states are plotted in Figs. 13 and 14, respectively, to understand the characteristics of change in pore structure in each pore size range. In terms of equivalent radius, micropores are defined as <100 μm; mesopores as 100–500 μm, macropores as 500–1000 μm, and extra-large pores as >1000 μm. The proportions and volume fractions of these pores are shown in Tables 6 and 7, respectively.

The data in Table 6 show that after F-T cycling, the proportions of mesopores in Specimens B and C decrease by 11.37% and 19.44%, respectively. In comparison, the proportions of micropores increase by 6.42% and 10.61% and those of macropores increase by 79.25% and 111.86%. Under load, due to the closure or inter-penetration of microfractures, the numbers of micropores and mesopores gradually decrease. Overall, however, the number of micropores decreases more and the proportion of mesopores increases. In contrast, the proportion of macropores increases before the peak stress point and decreases afterwards due to a small number of extra-large pores forming after the rock bridge is penetrated.

Figs. 13 and 14 show that under F-T cycling, the peak volume fraction of Specimen B decreases gradually but remains in the range of 300–350 μm, while that of Specimen C first decreases and then increases, with the range shifting from 300–350 μm to 350–400 μm. The load causes extra-large pores to appear so that the volume fraction curve changes from having a single peak to having multiple peaks. It can be seen from Table 7 that throughout the process, the volume fractions of micropores and mesopores tend to decrease, with those in Specimen B decreasing by 42.44% and 66.77% and those of Specimen C decreasing by 54.23% and 61.20%, respectively. The volume fraction of macropores gradually increases during the F-T process and before the peak stress point. After that point, the specimen is penetrated and the macropores combine to form extra-large pores, so their volume fraction dominates in the residual strength stage. It is worth noting that although there are a large number of micropores, their volume fraction is only about 1%. The number of mesopores decreases gradually during F-T cycling but their volume fraction accounts for >70%. Deterioration due to F-T cycling is also manifested as a significant increase in the volume fraction of macropores, making penetration failure of the rock mass after loading more likely.

4. Discussion

4.1. Mechanism of cracking in ice-filled fractures

No single mechanism is responsible for the deterioration of rock mass properties in F-T environments. The mechanisms involved may include volumetric expansion (Ishikawa et al., 2004), hydrostatic pressure (Powers, 1945), disjoining pressure (Derjaguin and Churaev, 1978), crystallisation pressure (Scherer, 1999) and solvent effects (Wang et al., 2016). However, certain mechanisms may dominate

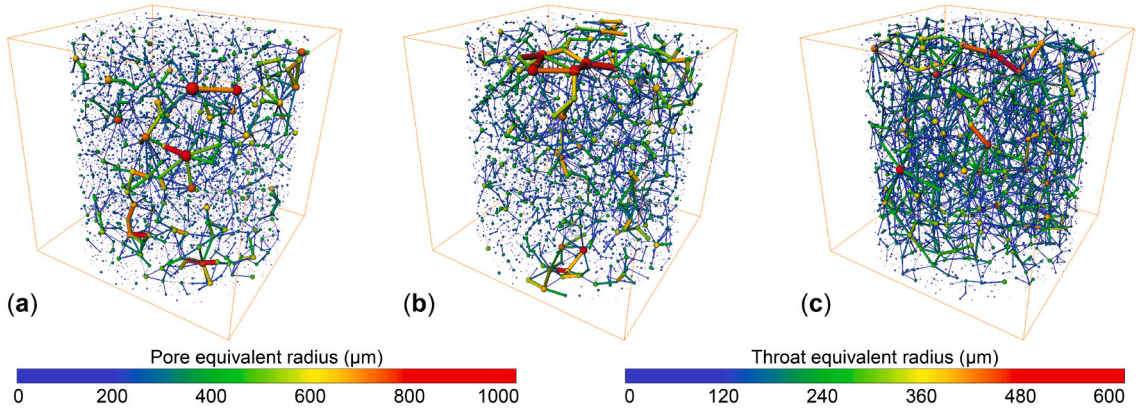


Fig. 12. PNMs of Specimen A after 50 F-T cycles (a), Specimen B after 30 F-T cycles (b), and Specimen C after 20 F-T cycles (c).

Table 4
Pore and throat parameters of Specimen B under various numbers of F-T cycles and stress states.

State	Pore equivalent radius (μm)		Average volume ($\times 10^6$ μm ^{3})	Number of pores	Throat equivalent radius (μm)		Throat length (μm)		Number of throats
	Max	Average			Max	Average	Max	Average	
0 F-T cycles	729.25	117.94	32.02	7218	383.15	95.47	2894.62	982.34	1136
15 F-Tcycles	901.53	112.77	32.36	7361	475.56	98.76	3414.42	1025.52	1275
30 F-T cycles	955.27	112.69	35.14	7607	560.85	102.20	3376.30	1048.44	1403
Peak stress	1063.25	112.15	43.23	7095	720.88	143.50	3701.17	1230.54	1153
Residual strength	1662.36	108.50	59.68	6695	1199.39	186.56	5105.89	1349.98	868

Table 5
Pore and throat parameters of Specimen C under various numbers of F-T cycles and stress states.

State	Pore equivalent radius (μm)		Average volume ($\times 10^6$ μm ^{3})	Number of pores	Throat equivalent radius (μm)		Throat length (μm)		Number of throats
	Max	Average			Max	Average	Max	Average	
0 F-Tcycles	757.67	116.24	32.57	7256	512.13	96.98	3130.34	1031.95	1046
15 F-T cycles	843.63	112.14	34.34	7558	526.15	88.05	2856.91	1013.31	1640
30 F-T cycles	919.99	112.19	39.91	7913	508.81	99.87	3733.27	1193.36	2508
Peak stress	1122.17	119.84	52.22	6977	880.59	116.83	4065.06	1239.20	2117
Residual strength	1741.84	116.78	72.36	6228	1280.51	156.82	5022.22	1326.30	1233

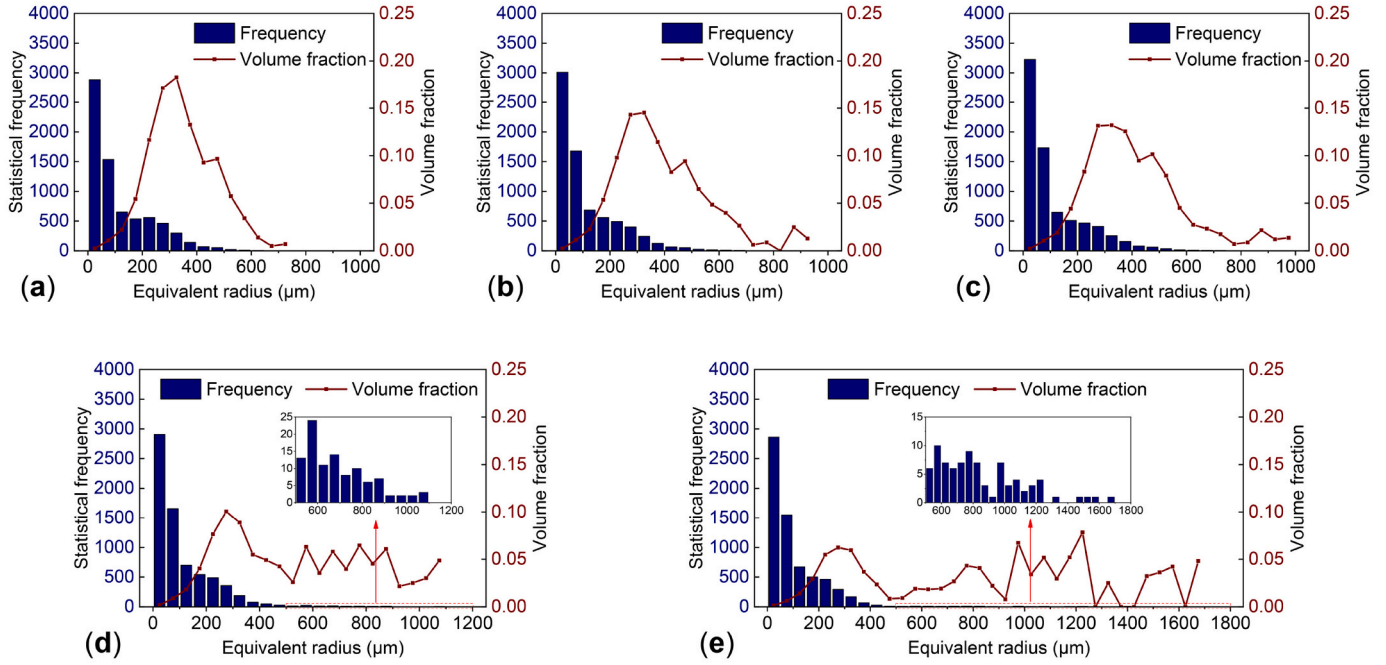


Fig. 13. Distributions of pore equivalent radius and pore volume fraction in Specimen B after 0 F-T cycles (a); 15 F-T cycles (b); 30 F-T cycles (c); and at peak stress (d); residual strength (e).

depending on different F-T conditions. The conditions and specimen size used in the present study reflect a rapid F-T environment, so volumetric expansion and hydrostatic pressure caused by water-ice phase transformation were the main factors leading to rock deterioration (Zhou et al., 2020).

The ice-filled fracture cracking mechanism is shown in Fig. 15. Under the influence of a temperature gradient, the freezing front and prefabricated fracture are perpendicular to each other, so the fracture freezes from the open end, which rapidly forms an ice-plug effect. Since the inside of the specimen is fully saturated with water, the volumetric expansion caused by phase transformation not only forms frost-heaving pressure at both ends of the fracture but also causes the unfrozen water to gather in the middle of the fracture. Also, because there is not enough free space in the closed fracture, hydrostatic pressure causes expansion of the prefabricated fracture, such that the original weak plane induces cracking of the secondary fracture. The 3D images in Fig. 6(c) and the 2D slices in Fig. 7(c) show that the deterioration in the middle part of the ice-filled F-T specimen is more severe near the end of the prefabricated fracture. In these two deterioration mechanisms, frost-heaving pressure plays a leading role in the short term. When the ice-plug effect occurs, increased hydrostatic pressure caused by free water being squeezed in the middle of the fracture is the root cause of fracture cracking (McGreevy and Whalley, 1985).

Furthermore, as unfrozen water is pressed into secondary fissures, the hydrostatic pressure gradually dissipates and makes it difficult for new fractures to expand in a non-open system. However, since F-T damage is irreversible, subsequent F-T cycles will continue to cause cracking at the tip and middle of the fracture, creating lower-grade micro-fractures.

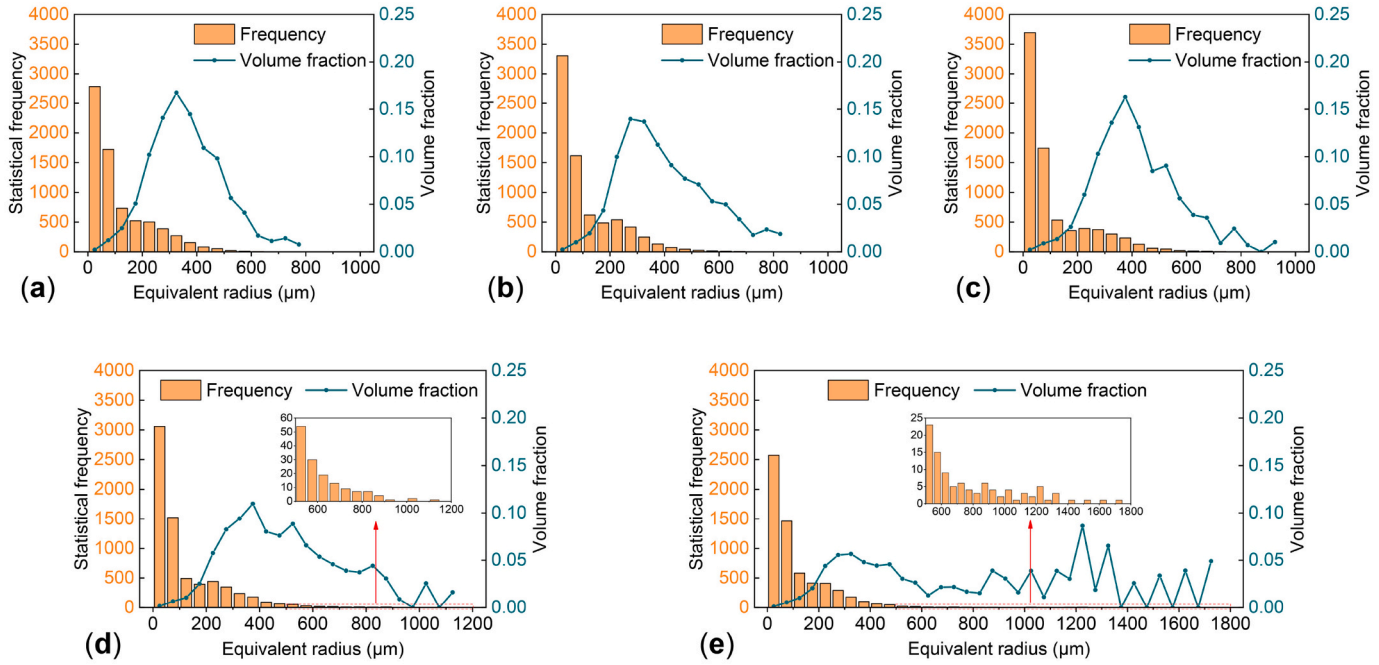


Fig. 14. Distributions of pore equivalent radius and pore volume fraction in Specimen C after 0 F-T cycles (a); 10 F-T cycles (b); 20 F-T cycles (c); and at peak stress (d); residual strength (e).

Table 6
Proportions of various pore types (%) in Specimens B and C.

Type	Specimen B					Specimen C				
	0 F-T cycles	15 F-T cycles	30 F-T cycles	Peak stress	Residual strength	0 F-T cycles	10 F-T cycles	20 F-T cycles	Peak stress	Residual strength
Micropores	61.22	63.65	65.15	64.33	65.89	62.12	65.10	68.71	65.62	64.83
Mesopores	38.27	35.54	33.92	34.24	32.86	37.29	33.90	30.04	32.28	33.56
Macropores	0.51	0.81	0.93	1.36	0.94	0.59	1.00	1.25	2.06	1.24
Extra-large pores	0	0	0	0.07	0.31	0	0	0	0.04	0.37

Table 7
Volume fractions of various pore types (%) in Specimens B and C.

Type	Specimen B					Specimen C				
	0 F-T cycles	15 F-T cycles	30 F-T cycles	Peak stress	Residual strength	0 F-T cycles	10 F-T cycles	20 F-T cycles	Peak stress	Residual strength
Micropores	1.33	1.39	1.29	1.06	0.77	1.42	1.23	1.09	0.82	0.65
Mesopores	86.91	75.37	73.19	47.09	28.88	83.77	71.97	71.71	53.61	32.50
Macropores	11.76	23.24	25.52	43.97	27.38	14.81	26.80	27.20	41.38	23.12
Extra-large pores	0	0	0	7.88	42.97	0	0	0	4.19	43.73

Additionally, although rapid freezing inhibits free water migration in line with the unfrozen water film to the frozen area, this mechanism still plays some role in causing slow frost-heaving damage to high-grade main fractures.

Under the action of hydrostatic pressure, ice-filled fractures are more likely to cause the expansion of rock mass structural planes. Therefore, the increase in S_{VF} in Specimen C after F-T cycling was as high as 76.11%. Meanwhile, the hydrostatic pressure under saturated F-T conditions was relatively low and was insufficient to drive a large number of fractures, so the increase in S_{VF} in Specimen B was only 35.08%.

4.2. Characteristics of fracture evolution

Existing research shows that F-T cycling has significant effects on rock mesostructures (Sun et al., 2020; Deprez et al., 2020b). In this section, the evolutionary characteristics of sandstone fracture structures during F-T cycling and after loading are discussed (Fig. 16).

According to the results, P_{tot} showed a trend of accelerated growth in the whole process of F-T, while the increase in P_R was far greater in the early F-T period than in the late period. Thus, hydrostatic pressure increased throat development mainly in the early F-T period, leading to

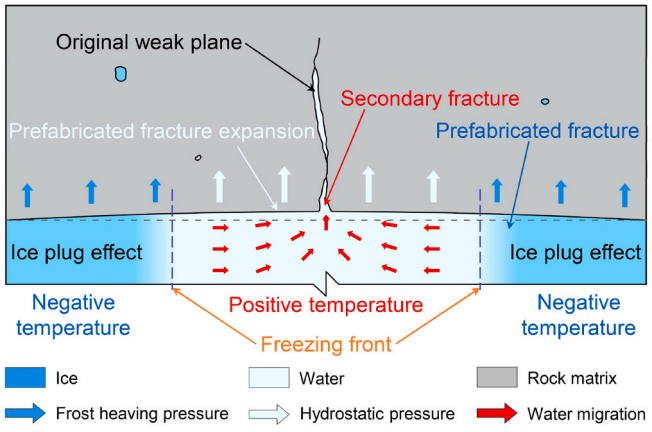


Fig. 15. Mechanism of cracking in an ice-filled fracture.

greater connection between the original pores and fractures. In the later F-T period, due to an increase in S_R, external moisture filled the previously isolated pores and fractures during thawing and increased the frost-heave pressure, which is manifested as accelerated rock damage (Fig. 16). These observations are consistent with those of Maji and Murton (2020), who conducted CT scanning of chalk under F-T cycling. Also, Table 5 shows that the average throat equivalent radius and throat length in Specimen C decreased at first and then increased. This further shows that the early F-T period mainly caused a sharp increase in the number of throats, which caused a sharp decrease in the number of non-connected pores.

The F-T effect causes many secondary fractures to become evenly distributed in the rock mass, so the U_z-value of the saturated intact specimen decreased gradually. In the fractured specimen, because F-T damage first occurred around the prefabricated fracture, the inhomogeneity expanded continuously during the early period of F-T. In the later period, due to the expansion of connected pores, the damage in the middle part of the rock bridge area intensified and the inhomogeneity decreased. As a result, the U_z-value of the saturated fractured specimen increased rapidly from 23.78% to 37.04% and then slowly decreased to 35.59%. However, due to the high internal hydrostatic pressure in the ice-filled specimen, the microfractures expanded rapidly in the early F-T period and became widely distributed throughout the specimen in the later period (Fig. 7(c)). As a result, the U_z-value of Specimen C increased from 26.91% to 30.23% and then dropped sharply to 16.85%, with no sharp increases in U_z being observed.

Macroscopically, the F-T effect caused continuous increases in porosity while, microscopically, there were different changes in pores according to their size. The results in Table 6 show that the proportion of mesopores gradually decreased, presumably because hydrostatic pressure primarily initiated microfractures and increased the number of micropores. Meanwhile, frost-heaving pressure forced the original mesopores to expand and form macropores, so the volume fraction of macropores increased rapidly (Table 7). This trend slowed down in the later F-T period due to some micropores transforming into mesopores as damage increased. After loading, the macropores expanded rapidly and connected to form extra-large pores, while the micropores and mesopores became compacted and closed (Fig. 16). Also, sandstone is cemented by many particles such that under F-T conditions it readily undergoes grain spalling, which reducing the internal bonding forces so that the maximum equivalent radius of pores increases rapidly after loading (Table 5).

On the other hand, the increases in P_{tot} and P_R during F-T cycling in saturated specimens were similar; however, a significant difference in their S_R curves is evident in Fig. 10. As PNMs can only reflect overall changes in pore-throat parameters, existing methods have difficulty in revealing differences in the spatial distributions of F-T damage. Further research should model the 3D spatial distributions of F-T damage in rock masses.

5. Conclusions

This paper investigated the evolutionary characteristics of secondary fractures in saturated intact, saturated fractured, and ice-filled fractured sandstone specimens under various numbers of F-T cycles and stress states using CT technology. The main conclusions are:

(1) Fracture development comprises four main processes: frost-heave cracking, progressive expansion of damage, rock bridge penetration under load, and shear failure. The S_{VF}-value of the fractured specimen peaked at the peak stress point but that of the saturated fractured specimen only increased by 35.08% after F-T cycling. In comparison, in the ice-filled fractured specimen, the S_{VF}-value increased by 76.11%.

(2) Parameter P_R can reflect the influence of F-T cycling on the connectivity of the pore structure. The P_R-values of the saturated intact, saturated fractured, and ice-filled fractured specimens

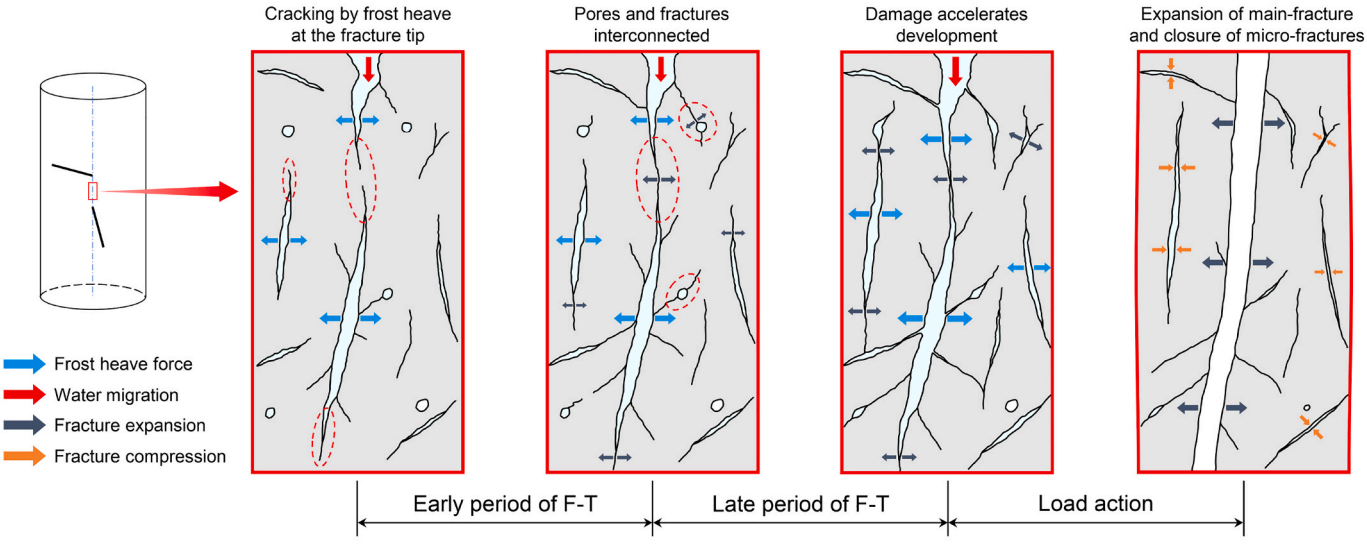


Fig. 16. Evolution in pores and fractures in sandstone under F-T cycling and uniaxial compression.

after F-T cycling were 67.54%, 60.20%, and 90.28%, respectively. The S_R curves can reflect differences in fracture expansion in different 2D layers under different F-T saturation modes.

(3) The effects of F-T cycling differ according to pore size. Micropores accounted for >60% of the number of pores and increased in number with F-T cycling more than pores of other sizes. The number of mesopores gradually decreased but their volume fraction always remained above 70%. The proportion and volume fraction of macropores increased the fastest. Therefore, F-T damage is manifested in two ways: 1) the number of micropores increases, causing the inside of the rock to become more broken, and 2) the volume of macropores increases, making the rock more susceptible to damage under load.

(4) Increases in hydrostatic pressure indirectly caused by water-ice phase transformation are the leading cause of cracking in ice-filled fractures. Therefore, greater changes in porosity, heterogeneity coefficient and pore-throat parameters occur in ice-filled rock than in saturated rock. Early F-T effects are mainly manifested as a sharp increase in the number of throats, while later effects are manifested as the expansion of pores and throats, which causes further damage.

The macroscopic deformation and failure of engineering rock masses involve a combination of micro-fracture initiation, expansion and penetration. Therefore, understanding slope failure and instability behaviours in cold regions depends on the in-depth study of meso-structural parameters. To reveal the internal mechanism of rock mass failure in cold regions, reasonable meso-damage variables should be selected in follow-up work to quantitatively describe meso-fracture evolution and macroscopic mechanical behaviour.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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